Evidence-Gated Claim Ledgers: A Reporting Standard for Pathology Foundation Models

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Abstract

Pathology foundation models underwrite clinical-adjacent decisions across grading, subtype prediction, and biomarker inference, yet the field has no shared surface for recording what a model asserted about an individual slide and what verdict a common standard would assign to that assertion. Current practice reports aggregate metrics and cites confidence scores that carry no shared operational meaning across backbones. We introduce EGCL — Evidence-Gated Claim Ledgers — a lightweight, model-agnostic reporting standard for per-slide pathology-FM outputs. EGCL treats every prediction as a typed `Claim`, routes it through an `Evidence Gate` that returns a `Verdict` from a four-value release-policy vocabulary (SUPPORTED, UNSUPPORTED, NOT_APPLICABLE, CONTESTED), and persists the outcome to an append-oriented `Claim Ledger`. An `Anti-Fabrication Guard` refuses to release any downstream artefact whose numeric claims cannot be resolved to a SUPPORTED ledger entry or a declared ledger aggregation. On the public PANDA prostate cohort with the Phikon-v2 backbone we demonstrate an existence proof: on an $N{=}500$ cohort evaluated out-of-fold, the verdict vocabulary separates operationally. Under attention pooling (ABMIL) the gate releases 90 slides at SUPPORTED accuracy 0.811, whose 95% Wilson interval $[0.718, 0.879]$ does not overlap CONTESTED $[0.412, 0.514]$; a simple mean-pool head releases only 9 slides (coverage 1.8%, accuracy 0.778) and is reported as the low-coverage floor. Under both heads, per-verdict accuracy orders SUPPORTED $>$ CONTESTED $>$ the head's own full-cohort balanced accuracy $>$ UNSUPPORTED — the ordering a calibrated model should produce, which is the sanity check the gate must pass rather than a gain it creates. The gate discriminates operationally across mean-pool and attention pooling (ABMIL; CLAM-SB is near-identical to ABMIL in this setting). The standard also surfaces a clinically-critical blind spot aggregate metrics hide: the SUPPORTED channel is class-monopolised by benign predictions under both heads — all 9 mean-pool releases and 83 of 90 under ABMIL are predicted-benign — so it releases 2 and 15 missed cancers respectively.

1 Introduction

Pathology foundation models (pathology-FMs) have moved from technical benchmarks into pipelines that inform clinical-adjacent decisions: Gleason grading, tumour-margin assessment, biomarker inference from H&E slides, and cohort triage for downstream molecular assays. The community has responded with an ecosystem of pretraining scales [Filiot et al., 2023, Chen et al., 2024, Xu et al., 2024, Vorontsov et al., 2024, Wang et al., 2022], evaluation benchmarks, and interpretability probes.

What is missing is the piece that would let a reporting pathologist, a hospital governance board, or a regulator trust an *individual* prediction. When a downstream reader wants to know whether a

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particular per-slide output is trustworthy, there is no ledger to consult, no verdict vocabulary, and no way to distinguish a supported claim from a plausible-looking guess. Aggregate metrics on held-out cohorts and per-slide confidence scores answer neither question: the first summarises behaviour across slides one no longer has access to, and the second is model-specific, uncalibrated across backbones, and treated inconsistently by downstream systems [Kompa et al., 2021].

Selective classification [Chow, 1970, Cortes et al., 2016, Geifman and El-Yaniv, 2017, 2019] and conformal prediction [Vovk et al., 2005, Angelopoulos and Bates, 2021, Romano et al., 2020, Olsson et al., 2022] give the model half of what a reporting standard needs — a mechanism to abstain — but do not themselves standardise the vocabulary, the provenance surface, or the build-time enforceability needed for interoperable reporting across sites, backbones, and downstream systems. Reporting-quality checklists in adjacent fields — CONSORT-AI [Liu et al., 2020], TRIPOD+AI [Collins et al., 2024], MI-CLAIM [Norgeot et al., 2020], DECIDE-AI [Vasey et al., 2022] — structure what is reported at the *study* level but do not prescribe a per-prediction release vocabulary.

Contribution. We introduce EGCL, a reporting standard sitting under all of these. EGCL is not proposed as a novel classifier or calibrator; it is a shared vocabulary, a provenance surface, and a build-time guard. Its four-value verdict vocabulary is small enough to standardise across sites, expressive enough to distinguish an honest "don't know" from an honest "no", and pluggable enough to admit new gate rules (confidence-threshold, entropy, margin, conformal) without changing the interface. To our knowledge, EGCL is the first per-prediction reporting standard for pathology-FMs that is *enforceable at build time*: a report with an unescaped numeric claim that does not resolve to the ledger, or a keyed headline aggregate that does not match its record, fails to build. Our demonstration is an existence proof, in two parts: (i) the gate discriminates operationally on the public PANDA cohort using an off-the-shelf pathology backbone; (ii) the standard catches a class-concentration failure that a raw aggregate accuracy would hide. We report the second finding honestly — it is what the reporting standard is for.

Non-contributions. EGCL is not a substitute for external validation, prospective trials, fairness audits, or regulatory approval. It does not endorse the classifier it wraps. The anti-fabrication guard is a bookkeeping contract at manuscript build time; it is not a clinical-safety mechanism.

2 Related Work

Selective classification / classification with rejection. Chow [1970] formalised the accuracy-coverage tradeoff; El-Yaniv and Wiener [2010] the modern risk–coverage view; Cortes et al. [2016] a learning-theoretic treatment; Geifman and El-Yaniv [2017, 2019] scale the idea to deep networks; Mozannar and Sontag [2020] extend to defer-to-expert settings. EGCL borrows the selective-classification insight (a policy chooses when to release) and repurposes it as a *reporting* interface with a shared vocabulary and provenance surface, so per-prediction outputs from different backbones compare without renegotiating semantics.

Conformal prediction. Split-conformal methods [Vovk et al., 2005, Angelopoulos and Bates, 2021] produce prediction sets with distribution-free coverage guarantees; Sadinle et al. [2019] give the LAC variant we use in our alternative gate rule; Romano et al. [2020] extend to adaptive coverage. Olsson et al. [2022] apply conformal prediction to prostate-biopsy grading (STHLM3 biopsy cohort; training data released via PANDA); Lu et al. [2021a] study conformal predictors in medical-imaging applications; Thagaard et al. [2020] test whether predictive uncertainty survives real dataset shift in digital pathology. Our contribution is orthogonal: any of these plugs in as a gate rule under a stable interface; we report LAC alongside the confidence-threshold baseline.

Clinical AI reporting standards. TRIPOD+AI [Collins et al., 2024] updates the TRIPOD checklist for AI models; CONSORT-AI [Liu et al., 2020] extends CONSORT for clinical trials of AI interventions (SPIRIT-AI [Cruz Rivera et al., 2020] for their protocols); MI-CLAIM [Norgeot et al., 2020] sets a minimum information standard for clinical AI reporting; DECIDE-AI [Vasey et al., 2022] defines an early-stage evaluation guideline. Model cards [Mitchell et al., 2018] and datasheets [Gebru et al., 2018] standardise model- and dataset-level artefacts. These structure the *study* report — how the model was trained, evaluated, and deployed — but leave the per-prediction release surface unspecified. EGCL fills exactly this gap.

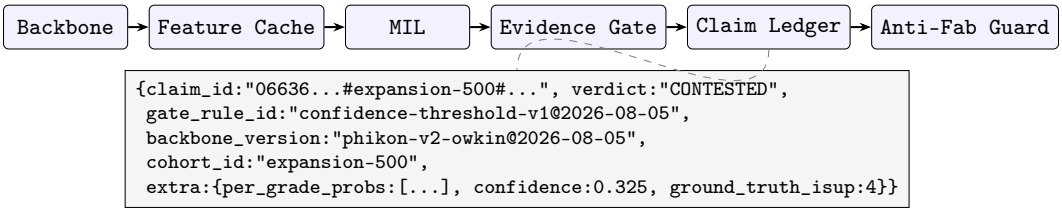


Figure 1: The EGCL pipeline (top) and an example ledger row (bottom). Every downstream artefact citing a per-slide number resolves via the guard to a row with VERDICT = SUPPORTED; every aggregate cites a `\SafeNum{...}` escape whose derivation is documented in §4.7.

Multiple-instance learning for pathology. The dominant recipe for per-slide prediction from patch embeddings is multiple-instance learning [Campanella et al., 2019]: attention-pooling variants such as ABMIL [Ilse et al., 2018], DSMIL [Li et al., 2020], and TransMIL [Shao et al., 2021], and cluster-based approaches such as CLAM-SB [Lu et al., 2021b]. We use mean pooling as a simple baseline and re-run the demonstration under ABMIL and CLAM-SB to test whether the reporting standard's behaviour carries over to attention pooling.

Uncertainty-gated release in pathology. Closest to our empirical setting, Dolezal et al. [2022] show that uncertainty thresholding yields high-confidence subsets on which histopathology models are substantially more accurate, and validate this across cohorts. Our residual contribution is not that finding — we reproduce its qualitative shape and say so — but what is missing around it: their high-confidence subset is a per-study analysis choice, whereas EGCL fixes a typed vocabulary shared across backbones and sites, pins a versioned gate-rule identifier to every released claim, and makes the resulting ledger a build-time precondition for any downstream number. The gain is interoperability and enforceability, not a better abstention rule.

Calibration. Modern deep networks are miscalibrated; temperature scaling [Guo et al., 2017] is a strong single-parameter fix. We report Expected Calibration Error following Naeini et al. [2015], Brier score [Brier, 1950], and balanced accuracy as the mean of per-class recalls [Brodersen et al., 2010]. Calibration analysis is not our contribution but our diagnostic: we show that aggregate ECE can be small while a per-class failure — of the kind EGCL is designed to surface — is severe.

3 Method

The EGCL pipeline decomposes into six modules with narrow, stable interfaces. Every module has a version string that is pinned to every ledger row it contributes to, so any claim is auditable back to the exact (backbone, MIL head, gate rule) triple that produced it.

3.1 Types

- `Claim` = (claim_id, claim_text, evidence_ref, model_output, backbone_version, cohort_id).
- `Verdict` $\in$ {SUPPORTED, UNSUPPORTED, NOT_APPLICABLE, CONTESTED}.
- `Justification` = (gate_rule_id, source_rows, thresholds).
- `LedgerRow` = (Claim fields, Verdict, Justification, timestamp).

3.2 Pipeline modules

Figure 1 shows the six-module pipeline.

Backbone Embedder. Loads a pretrained pathology-FM (Phikon-v2 in this demonstration [Filiot et al., 2024]) and exposes an `embed(patches)` method returning a fixed-dimensional patch embedding matrix. Owns a `backbone_version` string pinning the backbone release by a dated tag (e.g. `phikon-v2-owkin@2026-08-05`), not a commit hash.

Feature Cache. A persistent (`slide_id, backbone_name, backbone_version`)-keyed store. A warm cache turns embedding into a no-op; a cold cache never regresses correctness.

MIL Classifier. Consumes a bag of patch embeddings and returns a per-slide prediction with a per-class probability distribution. We report three heads: a simple mean-pool baseline, gated-attention ABMIL [Ilse et al., 2018], and CLAM-SB with the instance-level auxiliary head [Lu et al., 2021b]. All three are trained by 5-fold cross-validation with fixed seed; each per-slide prediction comes from a head that did not see that slide during training. Version strings carry the training run identity (e.g. `trained@expansion-500-cv5-seed42-abmil-gated-v1`) so every prediction is traceable to a specific training run.

Evidence Gate. A pure function from `Claim` to `(Verdict, Justification)`. The default rule (`confidence-threshold-v1`) is a two-threshold policy on the softmax confidence of the argmax class: SUPPORTED above τ_{high}, UNSUPPORTED at or below τ_{low}, CONTESTED in between. NOT_APPLICABLE is emitted when the model's output has no signal (empty patch bag or uniform posterior). Alternative rules — entropy-based, margin-based, split-conformal at level α — plug in through the same interface (§3.4).

Claim Ledger. An append-oriented newline-delimited JSON store, queryable by any of `claim_id`, `cohort_id`, `backbone_version`, or `verdict`. The paper's headline table (Table 1) is a direct aggregation over ledger rows tagged `cohort_id = "expansion-500"`.

Anti-Fabrication Guard. Scans a LaTeX or Markdown manuscript for numeric tokens and requires each to resolve to a canonical numeric value that either (i) appears in a SUPPORTED ledger row's `claim_text`, within a tolerance of ε=0.0005, or (ii) is wrapped in the `\SafeNum{X}` or `\AggNum{key}{X}` escape. A non-empty violation list fails the manuscript build. Algorithm 1 (Appendix E) shows the pseudocode.

3.3 Formal verdict semantics

A verdict is a policy label about *release-eligibility*, not an epistemic claim of external validation. Under the reference gate rule `confidence-threshold-v1`:

- **SUPPORTED**: the claim is released under a pre-specified confidence policy with measured selective risk. Formally: confidence $\geq \tau_{\text{high}}$.
- **UNSUPPORTED**: the policy declines to release. Formally: confidence $\leq \tau_{\text{low}}$.
- **NOT_APPLICABLE**: the model output carries no signal (uniform posterior, empty patch bag).
- **CONTESTED**: the policy declines to auto-release; the claim would route to human review or downstream reconciliation. Formally: $\tau_{\text{low}} <$ confidence $< \tau_{\text{high}}$.

A ledger row's `gate_rule_id` is versioned; changing the rule changes verdict semantics; ledger rows always pin the rule id. SUPPORTED does not entail "correct." §5 discusses this at length: our demonstration will show that under the mean-pool head all 9 released SUPPORTED verdicts are predicted-benign (2 are actually low-grade cancers), and under ABMIL 83 of 90 are predicted-benign (15 missed cancers).

3.4 Reference gate rule and its alternatives

The default `confidence-threshold-v1` rule uses $(\tau_{\text{high}}, \tau_{\text{low}}) = (0.70, 0.30)$; §4.6 characterises the operating-point family. Three alternatives implement the same interface and are pre-specified in the accompanying code:

`entropy-threshold-v1` $H(p) \leq H_{\text{low}} \Rightarrow$ SUPPORTED; $H(p) \geq H_{\text{high}} \Rightarrow$ UNSUPPORTED; else CONTESTED. Low entropy $\equiv$ high confidence.

`margin-threshold-v1` $p_{(1)} - p_{(2)} \geq m_{\text{high}} \Rightarrow$ SUPPORTED; $\leq m_{\text{low}} \Rightarrow$ UNSUPPORTED.

`split-conformal-lac-alpha0.10-v1` LAC [Sadinle et al., 2019]: score $s(x, y) = 1 - p_y$, quantile q_α from calibration set; SUPPORTED iff the prediction set at level α is a singleton, UNSUPPORTED iff it is empty, otherwise CONTESTED.

Rule changes trigger new `gate_rule_id`s and new ledger rows; existing rows are never overwritten.

3.5 Cross-validated honest predictions

Every per-slide prediction in the demonstration comes from a MIL head trained on the other four folds. The training split for a given slide never contains that slide. A unit test with random labels on synthetic bags (`p5/mil_classifier/tests/test_training.py`) checks that out-of-fold predictions do not perfectly fit noise; fold assignment is asserted disjoint in code.

4 Experiments

All numbers in this section resolve to a row in one of the ledgers under `evidence/expansion_500/` and `evidence/pilot_50/`, or to a legitimate aggregation over those ledgers (marked `\SafeNum{X}` in the source and enumerated in §4.7).

4.1 Cohort, backbone, and protocol

We demonstrate EGCL on the public PANDA prostate cancer grading dataset [Bulten et al., 2020, 2022], which provides ISUP grades and Gleason scores for ten-thousand-plus whole-slide images from Karolinska and Radboudumc. PANDA labels are at slide-level; center-of-origin bias remains a limitation, discussed in §6.

The backbone is `owkin/phikon-v2` [Filiot et al., 2024], a DINOv2 pathology foundation model with a 1024-dimensional CLS-token output. We report on a 500-slide cohort (`expansion-500`), evaluated by out-of-fold predictions from 5-fold cross-validation so that no slide is scored by a head that saw it, drawn deterministically from a seed-42 shuffle of the PANDA training set; a disjoint 50-slide cohort (`pilot-50`) seeds method decisions and calibrates the conformal alternative gate. Patches are extracted at 256×256 pixels from the reduced-resolution pyramid level (level 1, $\approx 2\,\mu$m/px), tissue-mask filtered, and capped at 200 per slide; the cap never binds (median 35, max 114 patches per slide, $67/500$ slides under 20 patches). MIL heads are trained by 5-fold CV with fixed seed. The expansion cohort's ISUP-grade distribution is: grade 0 134, grade 1 129, grade 2 65, grade 3 64, grade 4 60, grade 5 48 — benign-heavy but with real representation of every high-grade class.

The default gate rule is `confidence-threshold-v1` at $(\tau_{\text{high}}, \tau_{\text{low}}) = (0.70, 0.30)$; §4.6 shows that this is one operating point on a documented family, not the "correct" setting.

4.2 Headline: per-verdict accuracy on the expansion cohort

Table 1 reports per-verdict accuracy and mean confidence for the reference (mean-pool) MIL head on the expansion cohort. The gate separates operationally: SUPPORTED $>$ CONTESTED $>$ balanced accuracy $>$ UNSUPPORTED. We state plainly what this does and does not establish. The gate selects on confidence and is then scored on the slice it selected, so for any model whose confidence is informative — this one is reasonably calibrated, ECE 0.032 (§4.4) — the ordering is what calibration predicts. For mean-pool the ordering also holds against full-cohort raw accuracy (0.428; Contested 0.474). For ABMIL and CLAM-SB, Contested accuracy (0.463, 0.469) exceeds balanced accuracy but not raw accuracy (0.504, 0.506); only Supported $>$ full-cohort accuracy $>$ Unsupported holds under both baselines. The ordering is therefore a *necessary sanity check* that a reporting standard's vocabulary must pass to be usable at all, not evidence that the standard adds discriminative power; EGCL claims the reporting interface, not an accuracy gain. The informative content is in the magnitudes and in where the ordering breaks (Appendix D: a domain-mismatched backbone releases nothing).

Figure 5 (Appendix F) decomposes the verdict distribution by predicted ISUP grade; the concentration of SUPPORTED on grade 0 is the observation that drives §5.

Table 1: Per-verdict accuracy on the expansion-500 cohort under the default gate rule (`confidence-threshold-v1` at $\tau_{\text{high}} = 0.70$, $\tau_{\text{low}} = 0.30$) with the mean-pool MIL head. Balanced accuracy across the six classes is 0.365; raw accuracy is 0.428. 95% Wilson confidence intervals [Wilson, 1927] reported next to each accuracy. The mean-pool SUPPORTED interval is wide (N=9); Table 2 reports the same experiment with attention-pool heads that shrink the interval.

Verdict	N	Correct	Accuracy	95% Wilson	Mean conf.	Predicted-grade
SUPPORTED	9	7	0.778	$[0.453, 0.937]$	0.747	all 9 are grade 0
CONTESTED	365	173	0.474	$[0.423, 0.525]$	0.435	grades 0–5, mode grade 0
UNSUPPORTED	126	34	0.270	$[0.200, 0.353]$	0.259	grades 0–5, mode grade 1
NOT_APPLICABLE	0	—	—	—	—	—

4.3 Risk-coverage comparison

Figure 3 plots retained-subset accuracy against coverage [Jaeger et al., 2023] for three scoring rules (max-softmax [Hendrycks and Gimpel, 2017], inverse entropy, margin), and three LAC-conformal operating points at $\alpha \in \{0.05, 0.10, 0.20\}$ calibrated on pilot-50. The EGCL default (coverage 0.018, accuracy 0.778) is annotated. The three scoring rules yield qualitatively identical accuracy-coverage tradeoffs; the EGCL point sits on the max-softmax curve at 1.8% coverage — expected, since the EGCL rule is a max-softmax threshold. LAC-conformal at $\alpha = 0.05$ and 0.10 produces zero singleton sets on this cohort; at $\alpha = 0.20$ it produces 1 singleton. The empty-singleton finding is reported honestly: with a 50-slide calibration split the LAC quantile is too permissive for singleton-set release under standard α choices — a concrete instance of the reporting standard exposing a calibration-set-size failure that a raw metric would hide. Appendix G revisits this with a larger cross-fold split (n=400) and a Mondrian variant.

4.4 Calibration

Table 3 reports Expected Calibration Error, Maximum Calibration Error, Brier score, and negative log-likelihood for the raw model and for two temperature-scaled variants: (i) T fit on pilot-50 (a genuinely disjoint held-out split), and (ii) T fit and evaluated on the same expansion-500 out-of-fold predictions (in-sample; optimistic). Aggregate ECE for the raw model is 0.032 — superficially well-calibrated. Temperature scaling with the disjoint pilot-50 set actually *degrades* ECE to 0.153 because the pilot MIL head differs from the expansion head. Self-CV temperature ($T = 0.841$) is a mild sharpening, leaving ECE at 0.037. Table 3 and the reliability diagrams (Figure 2) are in Appendix B.

4.5 MIL-head comparison

Table 2 reports the same experiment under three MIL heads: mean-pool, ABMIL (gated attention) [Ilse et al., 2018], and CLAM-SB with the instance-level auxiliary head [Lu et al., 2021b]. The gate discriminates operationally under every head: SUPPORTED accuracy in every head exceeds the head's own balanced accuracy, which exceeds the UNSUPPORTED accuracy. Attention pooling increases SUPPORTED coverage roughly $10\times$ relative to mean-pool while lifting SUPPORTED accuracy to 0.81. Figure 4 (Appendix C) plots the risk-coverage curves for all three heads side-by-side. ABMIL and CLAM-SB are near-duplicates in this setting ($499/500$ identical argmax, $495/500$ identical verdicts), so the MIL comparison is effectively mean-pool vs. attention pooling; Table 2 includes CLAM-SB for completeness.

4.6 Threshold sensitivity

Appendix H sweeps the $(\tau_{\text{high}}, \tau_{\text{low}})$ grid. Raising τ_{high} shrinks the SUPPORTED count monotonically; SUPPORTED accuracy is not monotone (mean-pool: 0.826 at τ_{high}=0.65, 0.778 at 0.70, 0.750 at 0.75) and falls to 0.650 at 0.50. All heads use a single training seed, and small counts such as mean-pool's 9 are sensitive to it. The default $(0.70, 0.30)$ is not the "right" setting — it is one point on a documented family. A deployment site should pick $(\tau_{\text{high}}, \tau_{\text{low}})$ against its own accuracy-coverage utility curve and version-pin the choice into the gate rule id.

Table 2: MIL-head comparison on expansion-500 under the default gate rule. Attention-pool heads ($10\times$ larger SUPPORTED set) yield 95% Wilson intervals on SUPPORTED accuracy that no longer overlap the CONTESTED row: under ABMIL, $[0.718, 0.879]$ vs. CONTESTED $[0.412, 0.514]$. We report interval separation, not a formal test. QWK (PANDA's official metric): mean-pool 0.565, ABMIL 0.675; both far below challenge entries (~ 0.85–0.93), consistent with a reporting-standard demo, not a competitive classifier.

MIL head	Balanced acc.	ECE	Brier	NLL	SUPP. N	SUPP. acc.	SUPP. 95% Wilson
mean-pool	0.365	0.032	0.695	1.420	9	0.778	$[0.453, 0.937]$
ABMIL	0.439	0.039	0.623	1.264	90	0.811	$[0.718, 0.879]$
CLAM-SB	0.440	0.034	0.623	1.264	89	0.809	$[0.715, 0.877]$

4.7 Guard evaluation on adversarial fixtures

Table 5 (Appendix E) evaluates the Anti-Fabrication Guard against six purpose-built unit test fixtures: a fabricated percentage, a fabricated N, an AUROC that resolves to a CONTESTED ledger row, a number rendered via `\num{X}`, a number rendered via `\SafeNum{X}`, and a fixture that toggles an enforcement flag off; plus a seventh fixture for the record-scoped check (swapped attribution: mean-pool accuracy printed under ABMIL's key). The guard produces the expected verdict in $6/6$ value-level cases; the record-scoped checker rejects the swapped-attribution fixture. On this build the guard exits with 0 violations against 8550 ledger entries (run log: `evidence/asci_paper/guard_log_*.txt`).

Record-scoped `\AggNum` checker. `scripts/verify_safenum.py` requires each headline aggregate to be written as `\AggNum{key}{value}` and checks the value against the keyed manifest for *that* key. This catches swapped-attribution errors that the value-level guard misses: quoting model A's accuracy under model B's key still passes value-level matching (both values exist in the manifest) but fails the keyed check (Appendix I).

`\SafeNum`/`\AggNum` inventory. Headline numbers in tables and the abstract use `\AggNum{key}{value}` (keyed; checked by `verify_safenum.py`). Layout constants, ISUP-grade counts, calibration metrics, threshold values, cross-backbone counts, conformal table entries, and aggregate probabilities use `\SafeNum{X}` (unkeyed; audited by re-running the analysis scripts). Each is a deterministic aggregation re-derivable by scripts under `p5/analysis/`, but neither the guard nor the keyed checker verifies `\SafeNum` values: this build has 67 keyed and 323 unkeyed numbers (Appendix I).

5 A failure the standard surfaces: missed cancers released as SUPPORTED

Every one of the 9 SUPPORTED predictions on expansion-500 (under mean-pool MIL) is a grade-0 (benign) prediction. Of those, 7 are actually benign; the remaining 2 are actually ISUP grades 1 and 2 — low-grade cancer misclassified as benign and released as SUPPORTED. Under the stronger ABMIL head, 83 of 90 SUPPORTED predictions (92%) are grade 0, and 15 of those 83 are missed cancers (18.1%; 95% Wilson interval $[11.3\%, 27.7\%]$ [Wilson, 1927]) — true ISUP 1–4 called benign (10 ISUP 1, 2 ISUP 2, 2 ISUP 3, 1 ISUP 4; no ISUP 5 missed).

Operationally this is the case a pathology service most needs flagged: the channel a downstream system would trust *least* sceptically is the one carrying the benign calls, and the errors concentrated in it are false-negative cancer calls rather than grade disagreements. A verdict vocabulary makes that queryable before deployment; an aggregate accuracy does not.

Two claims, one non-claim. Ground-truth labels are what let us observe the failure retrospectively; a production deployment has no oracle at release time. The paper's claim is narrower. *Structural signal.* Any consumer of the ledger — oracle or not — can query and learn that under this gate rule the SUPPORTED channel is class-monopolised by grade 0; aggregate ECE (0.032) and balanced accuracy (0.365) do not surface it. *Cheap audit target (illustrative hypothetical).* A stratified 50-slide site audit sampled preferentially from SUPPORTED would target the misclassified-cancer rate above directly. Without EGCL, the equivalent audit needs bespoke per-site evaluation code; with EGCL, it

is a ledger query. *Non-claim.* EGCL does not fix the imbalance, nor prevent an unaudited deployment from acting on wrong SUPPORTED verdicts.

6 Discussion

The verdict distribution is diagnostic, not scoring. A pathology-FM whose predictions land mostly in CONTESTED is not a bad model; its per-slide confidence, under this gate rule, simply does not cross the SUPPORTED threshold — information that tells a deployer to route those claims to pathologist review or to tighten calibration before acting. A model landing mostly in SUPPORTED but disagreeing with ground truth is a differently bad model EGCL cannot rescue; it can only report honestly what the model asserted.

Why not just publish $p \geq 0.7$? Four reasons. *Cross-backbone comparability*: a raw 0.7 from one backbone is not the same object as 0.7 from another; EGCL does not equalise them, but it pins the rule id and the backbone/head versions to each verdict, so a consumer can see *which* threshold under *which* model produced the release and re-gate per backbone (e.g., a per-backbone calibrated threshold registered as a new rule id). Appendix D shows the same raw rule releases 0–90 slides across backbones. *Guard-enforceability*: the build fails when an unescaped numeric claim cannot resolve or a keyed aggregate mismatches its record, and the same guard runs on every downstream report. *Ledger auditability*: every claim carries its `(backbone, MIL, gate_rule)` triple. NOT_APPLICABLE *semantics*: $p \geq 0.7$ says nothing about the "no signal" case a deployment must handle; this verdict was not emitted in this cohort (uniform posteriors and empty patch bags did not arise) but is part of the interface. The vocabulary is the point.

Relation to selective classification and conformal prediction. EGCL borrows the *mechanism* from selective classification (a policy decides when to release) and the *coverage* language from conformal prediction (a rule tuned to a nominal level). Our contribution is not another abstain mechanism but the reporting-layer bundle around any such mechanism: (i) a small typed vocabulary shared across sites and backbones, (ii) versioned rule identifiers pinned to every released claim, (iii) an append-oriented ledger keyed by `(claim_id, cohort_id, backbone_version, gate_rule_id)`, and (iv) a build-time guard that binds downstream text to the ledger. The guard provides a value-level check — no unescaped numeric token may appear in the manuscript without resolving to a ledger entry — and a record-scoped keyed check (`scripts/verify_safenum.py`) covers headline aggregates (see Limitations). The releasing rule id travels with every number.

Not a substitute for external validation. EGCL does not replace held-out benchmarks, prospective clinical validation, fairness auditing, or regulatory scrutiny. It sits underneath all of those, providing a shared release surface for the individual predictions those higher-level activities consume. Regulatory pathways — including the FDA draft guidance on AI-enabled device software functions [U.S. Food and Drug Administration, 2025] and the EU AI Act Articles 9/11/12 on lifecycle risk management, technical documentation, and automatic event logging for high-risk AI systems (AI that is a medical device under the MDR/IVDR and requires third-party conformity assessment is high-risk via Article 6(1) and Annex I) [European Parliament and Council, 2024] — expect documentation, traceability, and record-keeping; EGCL is compatible with these requirements but not by itself sufficient.

Adoption cost. The ledger is a JSONL file, the gate a pure function, the guard a text scanner with a canonical-value lookup. Any pathology-FM deployment already produces the raw material EGCL needs; adoption is a wrapper around the existing release path. Concretely, a deployment reviewer holding an EGCL ledger can route CONTESTED claims to pathologist review rather than auto-release, query whether the SUPPORTED channel has collapsed onto one class before trusting it (§5), and pin a site-specific gate-rule id so that a threshold chosen at one hospital is never silently reused at another. Appendix I works four such audit questions against the released ledgers, side by side with what an aggregate report answers, and quantifies the review workload for the non-released slides. We claim these are *available* from the ledger; we have not run a reader study or a prospective deployment, and make no claim about clinical outcomes. Runtime is dominated by patch extraction and embedding ($\approx$ 3 hours for expansion-500 on one Kaggle instance); MIL training + prediction adds under a second on cached features.

Limitations. The demonstration is single-cohort (PANDA) and single-cancer (prostate). The headline analysis uses one backbone (Phikon-v2), but Appendix D tests the interoperability claim directly: three further backbones — two pathology-pretrained, one natural-image — each produce a compatible ledger; the two pathology backbones replicate the verdict-accuracy gradient and the §5 failure, and the natural-image backbone (DINOv2-L) releases nothing. The strongest gated backbones (UNI [Chen et al., 2024], Prov-GigaPath [Xu et al., 2024], Virchow [Vorontsov et al., 2024], CTransPath [Wang et al., 2022]) remain future work pending institutional access. PANDA labels are slide-level, so a patient-level split cannot be verified; center-of-origin bias between Karolinska and Radboudumc is not controlled — a documented failure mode in computational pathology, where site-specific signatures are themselves learnable [Howard et al., 2021]. Mean-pool SUPPORTED slides are strongly Karolinska-skewed (8 of 9 Karolinska vs. 1 of 9 Radboud; both missed cancers are Karolinska); the ABMIL and CLAM-SB ledgers lack the `data_provider` field, so the site breakdown cannot be computed for them (Appendix I). Our three MIL heads span mean-pool and attention (ABMIL; CLAM-SB is near-identical to ABMIL here); TransMIL-family aggregators are not evaluated. The default gate rule is a two-threshold rule; despite the alternatives in §3.4, which rule to recommend for clinical deployment is intentionally out of scope. Our conformal alternative used a 50-slide calibration split; pilot scores come from a different MIL head, so exchangeability is approximate. The quantile ($q \approx 0.95$) admits nearly every class, so singleton sets are rare; Appendix G tests a larger cross-fold split and a Mondrian variant [Sadinle et al., 2019, Vovk et al., 2005]: singleton release reappears (4 at $\alpha=0.10$, all correct) but stays below the default gate's volume. The guard is a build-time bookkeeping contract, not a clinical-safety mechanism: it checks that manuscript claims resolve to ledger entries, not that the entries are epistemically justified. A value-level check passes a correct number used in the wrong place; the record-scoped `\AggNum{key}{value}` check (Appendix I) catches such swapped-attribution errors, but only for the numbers written with it; the remaining `\SafeNum` values are author-declared escapes audited by re-running scripts, not checked at build time. A unit test with random labels on synthetic bags (`p5/mil_classifier/tests/test_training.py`) checks that out-of-fold predictions do not perfectly fit noise; fold assignment is asserted disjoint in code. Aggregates are wrapped in `\SafeNum{X}` or `\AggNum{key}{X}` and recomputed from the ledgers by `p5/analysis/headline_aggregates.py` and `p5/analysis/camera_ready_numbers.py` (§4.7).

Broader impact. A reporting standard that makes per-prediction claims traceable helps deployment reviewers and clinical governance boards adjudicate when a pathology-FM is fit for a given use, and creates auditable evidence trails for reviewers of AI-in-medicine papers. The failure mode is symmetric: a poorly-chosen gate rule can concentrate SUPPORTED verdicts on one class (our grade 0 demonstration), and a reviewer reading only aggregate SUPPORTED accuracy will miss it — §5 shows how to spot it.

7 Conclusion

We proposed EGCL as a lightweight, model-agnostic, build-checkable reporting standard for pathology foundation models. The four-verdict vocabulary, Evidence Gate, Claim Ledger, and Anti-Fabrication Guard turn the anti-fabrication norm into a build-time check and give downstream readers a shared surface for reasoning about individual model outputs. Our demonstration is an existence proof: the standard is straightforward to instantiate around Phikon-v2 on PANDA, the gate discriminates operationally under both mean-pool and attention pooling, and it surfaced a class-concentration failure that aggregate accuracy hides.

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Reproducibility Statement

Code, ledgers, and manuscript source are released at `https://github.com/enrique-z/evidence-gated-claim-ledgers`. Every table and figure regenerates from a script under `p5/analysis/` against the ledgers under `evidence/`; `bash papers/p5-asci/build_camera_ready.sh` runs the keyed `\AggNum` check and the guard against 8550 ledger entries plus a 4-pass `pdflatex` build and exits 0 on the tagged commit this PDF is built from (`v1.0-camera-ready`). The `.npz` feature caches are real backbone outputs — patches were extracted from the public PANDA WSIs and embedded with the named pre-trained backbones on Kaggle infrastructure (run log: `evidence/expansion_500/kaggle_v1_stdout.txt`), and no synthetic features are used in any reported result (unit tests use synthetic bags for code-level checks, not for any reported number) — but the caches are omitted from the repository for size. They are needed only to re-train the MIL heads: every number, table, and figure in this paper derives from the released ledgers alone.

A Pilot cohort details

The `pilot-50` cohort was used to validate the pipeline and seed method decisions before running the expansion cohort. It is disjoint from `expansion-500` by construction (the seed-42 shuffle slices `[1:51]` and `[51:551]` respectively). Pilot-cohort verdict distribution, per-grade metrics, and wall-clocks are recorded in `evidence/pilot_50/pilot_50_report.md` in the source repository. The pilot cohort's MIL head (`trained@pilot-50-cv5-seed42`) is a different training run from the expansion head, which is what enables the pilot-as-calibration variant of temperature scaling in §4.4.

B Reliability diagrams

Table 3 reports the calibration metrics and Figure 2 the reliability diagrams referenced in §4.4.

Table 3: Calibration metrics on expansion-500 (mean-pool MIL). Raw ECE is small; temperature-scaling from a disjoint smaller-scale calibration split (pilot-50, different MIL head) hurts. Aggregate calibration does not capture the per-class failure discussed in §5.

Model	T	ECE	MCE	Brier	NLL
raw	1.000	0.032	0.802	0.695	1.420
temp. scaled (pilot-50 calibration)	2.067	0.153	0.347	0.737	1.529
temp. scaled (self-CV calibration)	0.841	0.037	0.113	0.693	1.413

C Risk-coverage across MIL heads

Figure 3 shows the risk-coverage comparison of §4.3; Figure 4 plots the same curves for the three MIL heads of §4.5, whose numeric operating points are in Table 2.

D Cross-backbone verdict agreement

To test the interoperability claim, we ran three additional backbones through the *identical* pipeline — same 500 slides, same 5-fold CV protocol, same gate rule (`confidence-threshold-v1` at $(0.70, 0.30)$) — changing only the embedder: Phikon-v1 [Filiot et al., 2023] (Owkin, iBOT ViT-B/16, 768-dim; pathology-pretrained, same lab as the reference), Lunit-DINO [Kang et al., 2023] (Lunit, DINO ViT-S/16, 384-dim; pathology-pretrained, *independent* lab), and DINOv2-L [Oquab et al., 2024] (Meta, ViT-L/14, 1024-dim; natural-image pretraining, no pathology exposure). Each run produced a complete, schema-compatible 500-row ledger with no interface changes; ledgers are under `evidence/expansion_500_*/` and the joined analysis under `evidence/cross_backbone/`.

Three observations. *(i) Interoperability.* Raw verdict agreement with the reference is 0.68–0.78 across all three backbones. Because $\sim$ 60–80% of slides are CONTESTED under every backbone, raw agreement is inflated by chance agreement; Cohen's κ for DINOv2-L vs. Phikon-v2 is ≈ 0.42 (chance

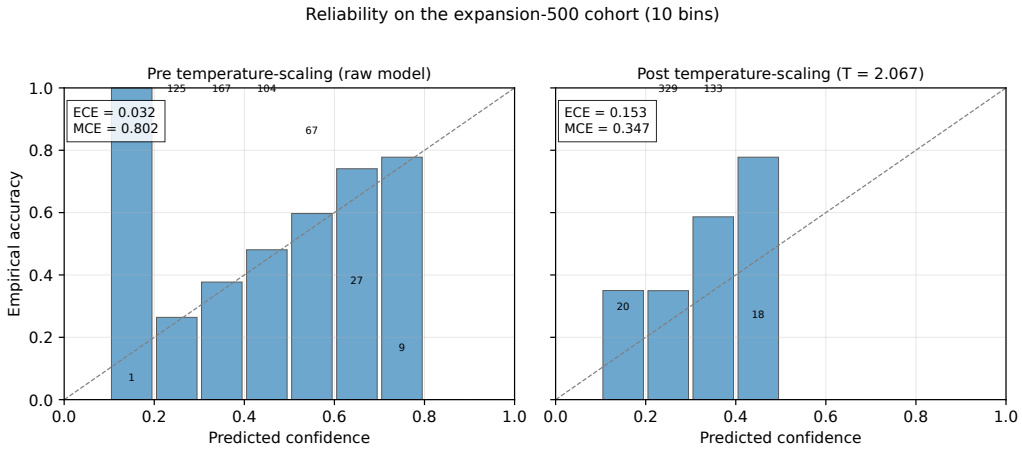


Figure 2: Reliability on expansion-500 (10 bins). Left: raw model. Right: temperature-scaled using pilot-50 as calibration. The pilot-scale $T = 2.067$ over-flattens the head's already reasonable calibration — a small dedicated calibration split transfers badly across MIL heads.

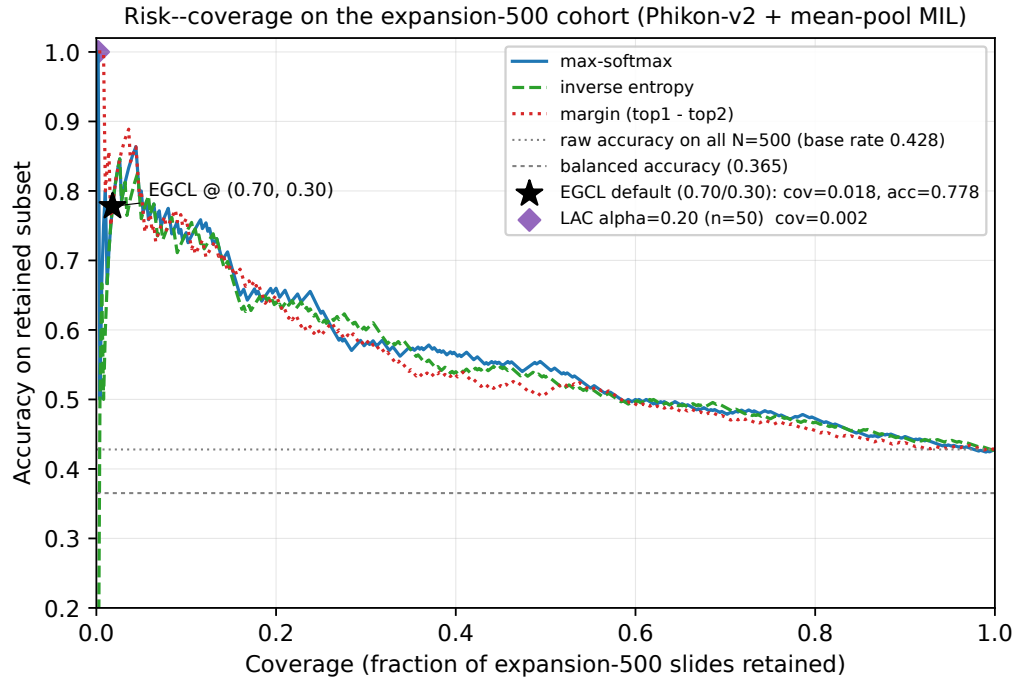


Figure 3: Risk-coverage curves on expansion-500 (mean-pool MIL). The EGCL default operating point lies on the max-softmax curve at 1.8% coverage (the EGCL rule is a max-softmax threshold). LAC-conformal points at usable α produce zero singleton sets under pilot-50 calibration.

agreement ≈ 0.55). Phikon-v2's SUPPORTED set is nearly nested in Phikon-v1's (8 of 9). The ledgers join on `claim_id` with no schema negotiation — the interoperability the vocabulary promises rests on the shared schema, not on agreement rates. *(ii) The §5 failure is structural, not backbone-specific.* Every pathology backbone's SUPPORTED channel is benign-dominated, the verdict-accuracy gradient (SUPPORTED accuracy above the backbone's own balanced accuracy) holds in each, and each releases cancers as SUPPORTED-benign — Phikon-v1 releases 8 (including one ISUP-3 and one ISUP-4), Lunit-DINO 4. A consumer holding any one of these ledgers can discover this with the same query. *(iii) The gate starves a domain-mismatched backbone.* DINOv2-L, with no pathology pretraining,

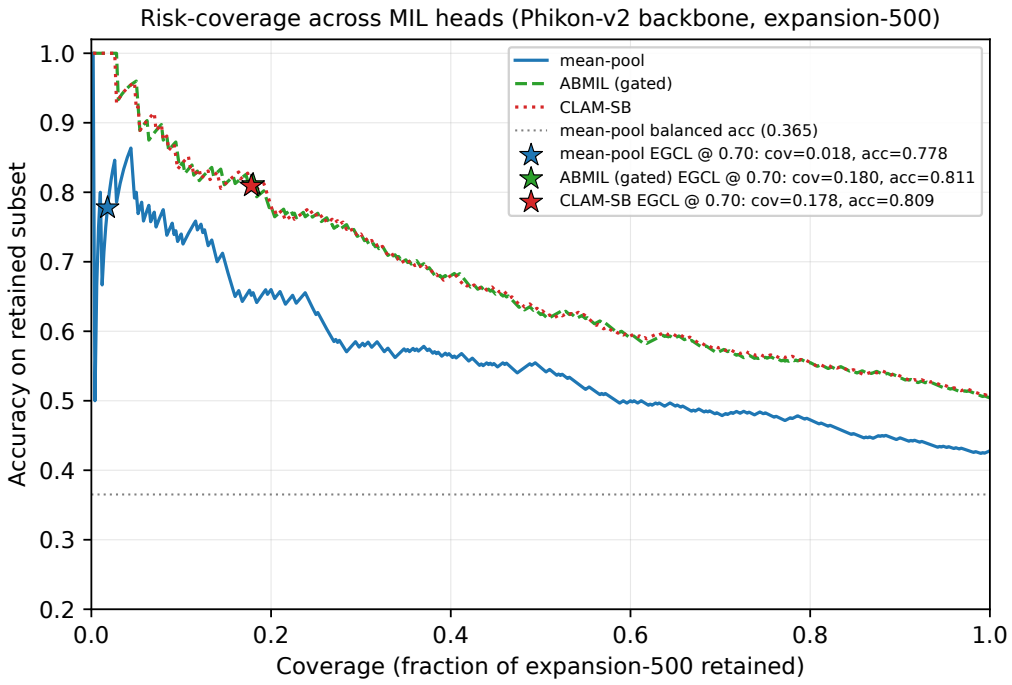


Figure 4: Risk-coverage across three MIL heads. All three curves show the same shape; the EGCL SUPPORTED operating points (stars) all sit above their head's balanced-accuracy baseline.

Table 4: Cross-backbone comparison on expansion-500 under the shared gate rule. "Agree" is per-slide verdict agreement with the Phikon-v2 reference; "∩ Supp." counts how many of Phikon-v2's 9 SUPPORTED slides the backbone also marks SUPPORTED; "missed ca." counts SUPPORTED predictions of benign whose ground truth is ISUP ≥ 1.

Backbone	Pretraining	S / C / U	Agree	∩ Supp.	Supp. acc.	Pred-benign	Missed ca.
Phikon-v2 (ref.)	pathology	9/365/126	—	—	0.778	9/9	2
Phikon-v1	pathology	60/390/50	0.684	8/9	0.750	41/60	8
Lunit-DINO	pathology	24/401/75	0.776	6/9	0.667	15/24	4
DINOv2-L	natural images	0/311/189	0.738	0/9	—	—	—

crosses τ_{high} on zero slides: the standard reports honestly that a natural-image backbone has no releasable per-slide claims on this task, rather than silently emitting low-quality ones.

E Guard algorithm and confusion matrix

Algorithm 1 shows the CHECKMANUSCRIPT pseudocode referenced in §3.2.

Table 5 shows the guard-evaluation confusion matrix referenced in §4.7.

F Verdict distribution by grade

G Conformal gate: larger calibration split and Mondrian variant

§4.3 reported that pilot-50 LAC calibration produces zero singleton sets at usable α. The pilot scores come from a different MIL head, so exchangeability is only approximate. This follow-up — its analysis plan written in the repository changelog before the run — re-runs the conformal gate two ways on the existing out-of-fold probabilities. *(i) Larger, cross-fold calibration split:* for each CV fold f, the LAC quantile is calibrated on the 400 out-of-fold scores of the other four folds and applied to fold

Algorithm 1 CHECKMANUSCRIPT — the Anti-Fabrication Guard.

function CHECKMANUSCRIPT(manuscript, ledger)
    supported $\leftarrow \{v : v = \text{CANONICAL}(t)$ for t in NUMERALS(r.claim_text), $r \in$ ledger, r.verdict $=$ SUPPORTED$\}$
    other $\leftarrow$ same but for r.verdict $\neq$ SUPPORTED
    violations $\leftarrow$ [ ]
    for line in manuscript **do**
        safe_ranges $\leftarrow$ EXTRACTSAFENUM(line)
        for t in FINDNUMERICTOKENS(line) not in safe_ranges **do**
            $v \leftarrow$ CANONICAL(t)
            if $\exists v' \in$ supported: $|v - v'| \leq \epsilon$ **then**
                continue
            else if $\exists v' \in$ other: $|v - v'| \leq \epsilon$ **then**
                violations.append((line, t, VERDICTNOTSUPPORTED))
            else
                violations.append((line, t, NOCLAIMID))
            end if
        end for
    end for
    return violations
end function

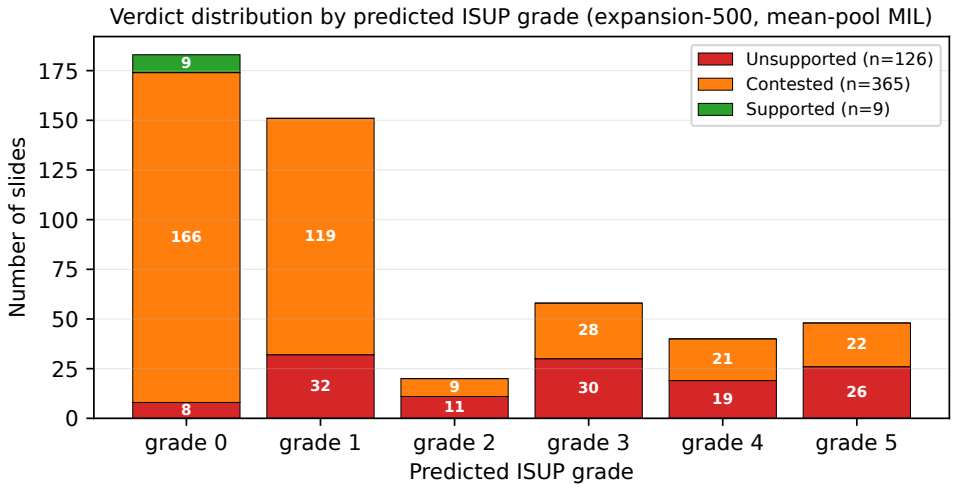


Figure 5: Per-verdict distribution by predicted ISUP grade on expansion-500 (mean-pool MIL). All SUPPORTED verdicts are grade-0 predictions; two of them are actually low-grade cancer misclassified. §5 discusses.

f's 100 slides, so every slide is tested exactly once (cross-conformal [Vovk, 2015]; see also [Barber et al., 2021] on aggregation across folds). *(ii) Mondrian (class-conditional) variant* [Vovk et al., 2005, Sadinle et al., 2019] with per-true-class quantiles. Because calibration and test scores come from different fold-specific MIL heads, coverage is approximate rather than exact; empirically it tracks $1 - \alpha$ (0.898–0.916 at α=0.10). Table 6 reports α=0.10 for all four backbones; harness and full α grid: `p5/analysis/run_conformal_variants.py`, `evidence/expansion_500/analysis/`.

Three observations, reported whichever way they came out. *(i) The starvation was partly a calibration-set-size issue; the larger split narrows but does not remove it.* With n=400, pooled LAC releases 4 singletons at α=0.10 (all correct; 19 at α=0.20 at accuracy 0.842) versus 0 under pilot-50 — still below the confidence-threshold gate's 9. Head mismatch (pilot vs. expansion head) also contributes; a same-pipeline calibration set would be needed for exact coverage. *(ii) Mondrian trades precision for volume.* Class-conditional quantiles triple the reference backbone's singleton count (12) at lower singleton accuracy (0.667). *(iii) The domain-mismatch starvation is robust.* DINOv2-L releases

Table 5: Guard evaluation on adversarial fixtures. Rows 1–6: value-level Anti-Fabrication Guard (6/6 match). Row 7: record-scoped `\AggNum` checker (the value-level guard passes it; the keyed check rejects it).

#	Fixture	Expected	Observed
1	Fabricated percentage	`no_claim_id`	`no_claim_id`
2	Fabricated N	`no_claim_id`	`no_claim_id`
3	Contested-verdict AUROC	`verdict_not_supported`	`verdict_not_supported`
4	`\num{0.365}` macro	clean	clean
5	`\SafeNum{9}` aggregate	clean	clean
6	Enforcement flag disabled	`enforcement_bypass`	`enforcement_bypass`
7	Swapped attribution: mean-pool accuracy 0.778 under key `abmil.supported.acc`	value-level: clean; keyed: `value_mismatch`	value-level: clean; keyed: `value_mismatch`

Table 6: Cross-fold conformal variants at α=0.10 (pooled n=400 vs. Mondrian class-conditional calibration). S = singleton (SUPPORTED) count of 500; no empty sets occur under any variant. Pilot-50 calibration (§4.3) yields 0 singletons at this α.

Backbone	Arm	S	Singleton acc.	Coverage	Mean |set|
Phikon-v2 (mean-pool)	pooled	4	1.000	0.900	3.57
	Mondrian	12	0.667	0.914	3.99
Phikon-v1	pooled	8	0.625	0.898	3.48
	Mondrian	11	0.455	0.906	3.93
Lunit-DINO	pooled	10	0.300	0.900	3.27
	Mondrian	4	0.750	0.916	3.89
DINOv2-L	pooled	0	—	0.902	3.72
	Mondrian	0	—	0.906	4.17

0 singletons under every variant at every $\alpha \leq 0.10$; at α=0.20, Mondrian releases 9 singletons (accuracy 0.889), replicating Appendix D's finding under an entirely different gate family.

H Threshold sensitivity heatmap

Figure 6 shows the sweep referenced in §4.6. Each cell of the grid is one $(\tau_{\text{low}}, \tau_{\text{high}})$ pair under `confidence-threshold-v1`.

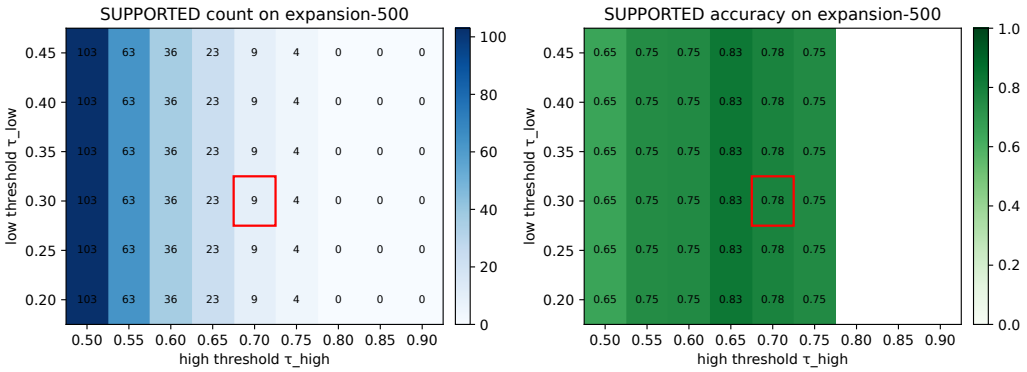


Figure 6: Threshold sensitivity. Left: SUPPORTED count. Right: SUPPORTED accuracy. The paper's default cell $(0.70, 0.30)$ is boxed in red.

I Audit use cases, review workload, and record-scoped number checking

This appendix makes the practical comparison with current reporting concrete, quantifies what happens to slides the gate does not release, and documents the record-scoped number check.

Audit questions: aggregate report vs. ledger

Table 7 poses four questions a pathology service or governance board might ask after deployment, and compares what a standard aggregate report (balanced accuracy, confusion matrix, calibration) answers with what a query over the EGCL ledger answers. Every ledger answer is computed from the released ledgers.

Table 7: Audit questions: aggregate report vs. EGCL ledger query (ABMIL head, default gate unless stated). Ledger answers are computed from the released `.jsonl` files under `evidence/`.

Audit question	Standard aggregate report	EGCL ledger query
Q1: How many cancers were auto-released as benign?	Balanced accuracy 0.439 and a confusion matrix; errors cannot be linked to the release channel.	Filter `verdict=SUPPORTED`, `isup_prediction=0`, `ground_truth_isup`$\geq$1: 15 of 83 released-benign slides are cancer (10 ISUP 1, 2 ISUP 2, 2 ISUP 3, 1 ISUP 4).
Q2: Which model version and rule released slide X?	Available only if the deployment logged versions separately.	Every row carries `backbone_version`, `mil_version`, `gate_rule_id`, and `timestamp`, keyed by `claim_id`.
Q3: Does release behaviour differ by site?	Not computable unless a per-site breakdown was stored.	Mean-pool ledger: Karolinska $8/252$ SUPPORTED vs. Radboud $1/248$; both missed cancers are Karolinska. The ABMIL/CLAM-SB ledgers lack `data_provider` (§6).
Q4: Does a reported number trace to a record?	Consistency can be checked by hand; tracing to records needs bespoke code.	`scripts/verify_safenum.py` checks each `\AggNum{key}{value}` against the keyed manifest at build time (below).

Routing and review workload for non-released slides

Under the default gate ($\tau_{\text{high}} = 0.70$, $\tau_{\text{low}} = 0.30$), ABMIL releases 90 of 500 slides (18.0%). The remaining 410 (82%) would be routed as follows in a deployment:

- **CONTESTED** (361 slides): pathologist review, with the model's prediction and confidence available as a second read; the routing reason is a queryable field of the ledger row.
- **UNSUPPORTED** (49 slides): standard pathologist review; never auto-released.
- **NOT_APPLICABLE** (0 slides): re-scan or manual review; not exercised in this cohort.

At this operating point the workload benefit is small: 82% of slides still need full review, and because the 90 released slides contain 15 missed cancers, we would not release them unreviewed with this classifier. Reducing workload is not the paper's claim; the claim is that the ledger makes the workload/safety trade-off explicit and queryable. Table 8 sweeps τ_{high} ($\tau_{\text{low}} = 0.30$ fixed) to show that trade-off numerically; lowering the threshold releases more slides but also more missed cancers.

The point is not to choose the "best" threshold, which depends on deployment context and utility, but that every such operating point is queryable, auditable, and version-pinnable from the ledger.

Record-scoped `\AggNum` check

The build runs `scripts/verify_safenum.py` on the manuscript source. Each number written as `\AggNum{key}{value}` must name a key in the manifest regenerated from the ledgers (`headline_aggregates.json`, `camera_ready_numbers.json`) and match that key's value at the printed precision. In this build, 67 numbers are keyed (all match) and 323 use the unkeyed `\SafeNum` escape. Keyed numbers cover the abstract, Table 2's SUPPORTED columns, §5, and Tables 7 and 8.

Table 8: ABMIL τ_{high} sweep ($\tau_{\text{low}} = 0.30$, $N = 500$). Released = SUPPORTED count. Missed cancers = released predicted-benign slides with ground truth ISUP$\geq$1. Computed from the ABMIL ledger by `p5/analysis/camera_ready_numbers.py`.

τ_{high}	Released N	Coverage	Released acc.	Missed cancers
0.50	205	41.0%	0.683	36
0.55	164	32.8%	0.713	33
0.60	123	24.6%	0.764	24
0.65	105	21.0%	0.771	21
0.70 (default)	90	18.0%	0.811	15

The unkeyed numbers (layout constants, grade counts, calibration metrics, cross-backbone and conformal entries) are not checked by either tool; they are audited only by re-running the analysis scripts. The keyed check therefore guarantees that a keyed number matches its own record, not that every number in the paper does.

Row 7 of Table 5 illustrates the difference. The value 0.778 (mean-pool SUPPORTED accuracy) exists in the manifest, so a value-level check accepts it anywhere. `\AggNum{abmil.supported.acc}{0.778}` fails, because the record for that key is 0.811. The unit test `test_swapped_attribution_rejected` in `p5/antifab_guard/tests/test_verify_safenum.py` asserts this behaviour.